



Hyperbolic Functions · Mark Scheme

Core Pure 2 Chapter 6: Hyperbolic Functions. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Definitions, graphs and identities

- 1 $\sinh(\ln 3) = \frac{1}{2} \left(3 - \frac{1}{3} \right) = \frac{4}{3}$ B1; $\tanh(\ln 3) = \frac{9-1}{9+1} = \frac{4}{5}$ B1.
- 2 $\cosh x$: U-shape, symmetric about the y-axis, through (0,1) B1; $\sinh x$: odd function through O, below $\cosh x$ and approaching it as $x \rightarrow \infty$ B1; range $\cosh x \geq 1$ B1.
- 3 $\cosh^2 x = \frac{1}{4} (e^{2x} + 2 + e^{-2x})$, $\sinh^2 x = \frac{1}{4} (e^{2x} - 2 + e^{-2x})$ M1; subtracting M1; $= \frac{1}{4} (4) = 1$ A1.
- 4 (a) $\cosh 2x \equiv 1 + 2\sinh^2 x$ B1. (b) $1 + 2\sinh^2 x = 1 + \frac{1}{2} (e^{2x} - 2 + e^{-2x})$ M1; $= \frac{1}{2} (e^{2x} + e^{-2x})$ M1; $= \cosh 2x$ A1.
- 5(a) RHS $= \frac{1}{4} [(e^A - e^{-A})(e^B + e^{-B}) + (e^A + e^{-A})(e^B - e^{-B})]$ M1; expanding, the e^{A-B} and e^{B-A} terms cancel M1; $= \frac{1}{4} (2e^{A+B} - 2e^{-(A+B)}) = \sinh(A+B)$ A1.
- 5(b) $\sinh 3x = \sinh 2x \cosh x + \cosh 2x \sinh x$ M1; $= 2\sinh x \cosh^2 x + (1 + 2\sinh^2 x) \sinh x = 2\sinh x (1 + \sinh^2 x) + \sinh x = 3\sinh x$ M1; $= 3\sinh x + 4\sinh^3 x$ A1.
- 5(c) $4\sinh^3 x - 8\sinh x = 0 \Rightarrow 4\sinh x (\sinh^2 x - 2) = 0$ M1; $\sinh x = 0$ or $\pm\sqrt{2}$ A1; $x = 0$, $x = \pm \ln(\sqrt{2} + \sqrt{3})$ A1.

Section B – Solving hyperbolic equations

- 6 $e^x - e^{-x} = \frac{3}{2} \Rightarrow 2e^{2x} - 3e^x - 2 = 0$ M1; $(2e^x + 1)(e^x - 2) = 0$, $e^x = 2$, $x = \ln 2$ A1.
- 7 $e^x + e^{-x} = \frac{5}{2} \Rightarrow 2e^{2x} - 5e^x + 2 = 0$ M1; $e^x = 2$ or $\frac{1}{2}$ A1; $x = \pm \ln 2$ A1.
- 8 $\frac{5}{2} (e^x + e^{-x}) - \frac{3}{2} (e^x - e^{-x}) = 5$ M1; $e^x + 4e^{-x} = 5 \Rightarrow e^{2x} - 5e^x + 4 = 0$ M1; $e^x = 1$ or 4 A1; $x = 0$ or $x = \ln 4$ A1.
- 9 $2\cosh^2 x - 1 - 5\cosh x + 4 = 0$ M1; $2\cosh^2 x - 5\cosh x + 3 = (2\cosh x - 3)(\cosh x - 1) = 0$ M1; $\cosh x = 1$ or $\frac{3}{2}$ A1; $x = 0$, $x = \pm \ln\left(\frac{3+\sqrt{5}}{2}\right)$ A1.
- 10(a) $2(e^x + e^{-x}) + \frac{1}{2}(e^x - e^{-x}) = k$ M1; $\frac{5}{2}e^x + \frac{3}{2}e^{-x} = k$ M1; multiply by $2e^x$: $5e^{2x} - 2ke^x + 3 = 0$ A1.
- 10(b) Real roots need $4k^2 - 60 \geq 0$ M1; $k^2 \geq 15$ A1; roots for e^x must be positive (product $\frac{3}{5}$, sum $\frac{2k}{5}$), so $k > 0$: $k \geq \sqrt{15}$ A1.
- 10(c) $5e^{2x} - 8e^x + 3 = (5e^x - 3)(e^x - 1) = 0$ M1; $e^x = 1$ or $\frac{3}{5}$ A1; $x = 0$ or $x = \ln \frac{3}{5}$ A1.

Section C – Inverse hyperbolic functions

- 11 $\operatorname{arsinh} 2 = \ln(2 + \sqrt{5})$ B1; $\operatorname{arcosh} 3 = \ln(3 + 2\sqrt{2})$ B1.
- 12 $x = \frac{1}{2} (e^y - e^{-y}) \Rightarrow e^{2y} - 2xe^y - 1 = 0$ M1; $e^y = x \pm \sqrt{x^2 + 1}$ M1; $e^y > 0$ so take + (since $x - \sqrt{x^2 + 1} < 0$), $y = \ln(x + \sqrt{x^2 + 1})$ A1.
- 13 $x = \tanh y = \frac{e^{2y} - 1}{e^{2y} + 1}$ M1; $xe^{2y} + x = e^{2y} - 1 \Rightarrow e^{2y} = \frac{1+x}{1-x}$ M1; $y = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$ A1.
- 14 $\operatorname{arsinh} \frac{3}{4} = \ln\left(\frac{3}{4} + \sqrt{\frac{25}{16}}\right)$ M1; $= \ln 2$ A1; $\operatorname{arcosh} x = \ln 4 \Rightarrow x = \cosh(\ln 4) = \frac{1}{2} \left(4 + \frac{1}{4} \right)$ M1; $x = \frac{17}{8}$ A1.
- 15(a) $x = \frac{1}{2} (e^y + e^{-y}) \Rightarrow e^{2y} - 2xe^y + 1 = 0$ M1; $e^y = x \pm \sqrt{x^2 - 1}$ M1 A1; $y \geq 0$ needs $e^y \geq 1$, but $x - \sqrt{x^2 - 1} = \frac{1}{x + \sqrt{x^2 - 1}} \leq 1$, so take +: $\operatorname{arcosh} x = \ln(x + \sqrt{x^2 - 1})$ A1.



15(b) Let $y = \operatorname{arcosh} x$; then $\cosh 2y = 17$, so $2x^2 - 1 = 17$ *M1*; $x^2 = 9$ *A1*; $x = 3$ ($x = -3$ rejected, as arcosh needs $x \geq 1$) *A1*.

15(c) $\operatorname{arcosh} 3 = \ln(3 + 2\sqrt{2})$ and $2 \operatorname{arsinh} 1 = 2 \ln(1 + \sqrt{2}) = \ln(1 + \sqrt{2})^2$ *M1*; $(1 + \sqrt{2})^2 = 3 + 2\sqrt{2}$, so they are equal *A1*.

Section D – Calculus of hyperbolic functions

16 (a) $4 \sinh 4x$ *B1*. (b) $2x \sinh x + x^2 \cosh x$ *B1*.

17 $\tanh x = \frac{\sinh x}{\cosh x}$; quotient rule: $\frac{\cosh x \cosh x - \sinh x \sinh x}{\cosh^2 x}$ *M1*; $= \frac{1}{\cosh^2 x}$ using $\cosh^2 x - \sinh^2 x = 1$ *M1*; $= \operatorname{sech}^2 x$ *A1*.

18 $\int \frac{1}{\sqrt{4x^2 - 9}} dx = \frac{1}{2} \int \frac{1}{\sqrt{x^2 - \frac{9}{4}}} dx$ *M1*; $= \frac{1}{2} \operatorname{arcosh} \left(\frac{2x}{3} \right) + c$ *M1 A1*. (Or $\frac{1}{2} \ln(2x + \sqrt{4x^2 - 9}) + c$.)

19 $x^2 - 2x + 5 = (x - 1)^2 + 4$ *M1*; $\int \frac{dx}{\sqrt{(x-1)^2 + 4}} = \operatorname{arsinh} \left(\frac{x-1}{2} \right)$ *M1*; $\left[\operatorname{arsinh} \left(\frac{x-1}{2} \right) \right]_1^3 = \operatorname{arsinh} 1 - \operatorname{arsinh} 0$ *A1*; $= \ln(1 + \sqrt{2})$ *A1*.

20(a) $dx = 2 \cosh u du$, $\sqrt{4 \sinh^2 u + 4} = 2 \cosh u$ *M1*; $\int 4 \cosh^2 u du$, limits 0 to $\operatorname{arsinh} 1$ *M1*; $= \int (2 + 2 \cosh 2u) du = [2u + \sinh 2u]$ *M1 A1*; $\sinh 2u = 2 \sinh u \cosh u = 2(1)(\sqrt{2})$ at the upper limit, giving $2 \ln(1 + \sqrt{2}) + 2\sqrt{2}$ *A1*.

20(b) $\int 1 \times \operatorname{arsinh} x dx = x \operatorname{arsinh} x - \int \frac{x}{\sqrt{1+x^2}} dx$ *M1*; $\int \frac{x}{\sqrt{1+x^2}} dx = \sqrt{1+x^2}$ *M1*; $x \operatorname{arsinh} x - \sqrt{1+x^2} + c$ *A1*.

Roar-level challenge

21 $\frac{dy}{dx} = 5 \sinh x - 3 \cosh x$ *M1*; $= 0 \Rightarrow \tanh x = \frac{3}{5}$ *M1*; $x = \operatorname{artanh} \frac{3}{5} = \frac{1}{2} \ln \left(\frac{1 + \frac{3}{5}}{1 - \frac{3}{5}} \right) = \frac{1}{2} \ln 4 = \ln 2$ *A1*; $\cosh(\ln 2) = \frac{5}{4}$,

$\sinh(\ln 2) = \frac{3}{4}$, so $y = \frac{25}{4} - \frac{9}{4} = 4$: $(\ln 2, 4)$ *A1*; $\frac{d^2y}{dx^2} = 5 \cosh x - 3 \sinh x = 4 > 0$, so a minimum *B1*.